Son Cao

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Education

Saxion University of Applied Sciences Enschede, Netherlands
B.Sc. Applied Computer Science *Aug 2024 – Jun 2028*

- **Focus:** Artificial Intelligence, Machine Learning, Embedded Systems, and Robotics.
- Project-based curriculum building end-to-end systems across software and hardware.

Experience

Saxion University of Applied Sciences Enschede, Netherlands
Applied Computer Science Student *2024 – Present*

- **AI/ML Coursework:** Built deep learning, computer vision, and classical ML pipelines as part of a project-based curriculum.
- **Embedded & Robotics:** Designed and implemented autonomous robots, real-time sensor systems, and TinyML deployments on microcontrollers.
- **Independent Projects:** Delivered end-to-end systems from data collection and model training to MCU deployment and live demonstrations.

Calvin Klein Ho Chi Minh City, Vietnam
Sales Assistant (Part-time) *2024*

- Provided high-quality customer service in a fast-paced international retail environment.
- Developed communication, teamwork, and time-management skills across diverse customer interactions.

Projects

Real-Time Hand Gesture Recognition
Python | TensorFlow/Keras | OpenCV | MediaPipe | CNN + Conv1D (TCN)

- **Hybrid Architecture:** CNN extracts spatial frame features; a Conv1D Temporal Convolutional Network models temporal gesture sequences.
- **Video Pipeline:** Built an OpenCV pipeline for frame extraction, normalization, and sliding-window sequence generation.
- **Custom Dataset:** Collected and trained on a 3-class dataset (thumbs up/down, stop); achieved high validation accuracy and real-time webcam deployment.

TinyML Air-Gesture Snake Game
C++ | Python | Edge Impulse | Arduino Nano 33 BLE | Pygame

- **On-device Inference:** Trained a gesture classifier with Edge Impulse and deployed it on Arduino Nano 33 BLE for TinyML inference.
- **Game Integration:** Built a Pygame Snake application that receives directional commands over serial from the microcontroller in real time.
- **Full Pipeline:** Data collection, model training, INT8 quantization, and MCU deployment with minimal latency.

Autonomous Maze Navigation Robot
*C++ | Raspberry Pi Pico | HC-SR04 | ITR20001 | PCA9685 | TB6612FNG | DFS | A**

- **Embedded System:** Built an autonomous maze-exploration robot with real-time 20×20 grid mapping on Raspberry Pi Pico.
- **Perception:** Combined ultrasonic (HC-SR04) and infrared (ITR20001) sensors for wall detection and environment perception.
- **Algorithms:** DFS with backtracking for maze discovery; A* pathfinding for computing the optimal goal-return path.
- **Actuation:** Precise motor control via TB6612FNG driver and PCA9685 PWM servo controller.

NVDA Stock Price Prediction
Python | scikit-learn | Pandas | Matplotlib | Alpaca API | Random Forest

- **Modelling:** Random Forest model predicting daily NVDA returns from historical market data (2022–2026).
- **Feature Engineering:** SMA, RSI, and volatility indicators with a time-aware train/test split for chronological integrity.
- **Results:** Achieved ~1.7% MAE; analyzed compounding error effects and model limitations on volatile time-series data.

Embedded Sensor Monitoring System
C / C++ | Raspberry Pi Pico | RTOS | Multiple Sensors | Serial Communication

- **Real-time Monitoring:** Multi-sensor system using RTOS task scheduling for concurrent data acquisition.
- **Reliability:** Live sensor readings displayed on a connected interface with robust edge-case and fault handling.

Hand-Controlled Snake Game
Python | Pygame | MediaPipe | Computer Vision

- **Computer Vision Control:** Built a classic Snake game controlled entirely by hand gestures captured via webcam using MediaPipe hand tracking.
- **Real-Time Interaction:** Counts raised fingers in real time to determine direction — no keyboard or mouse required.

Skills

- **Languages:** Python, C, C++, C#, JavaScript, HTML, CSS, SQL.
- **AI & Data Science:** Machine Learning, Deep Learning, Computer Vision, TensorFlow, Keras, scikit-learn, OpenCV, MediaPipe, Pandas, NumPy, Matplotlib.
- **Embedded Systems:** Raspberry Pi Pico, Arduino, ESP32, Arduino Nano 33 BLE, RTOS, Edge Impulse (TinyML), sensor integration, motor and servo control.
- **Tools & CS Concepts:** Git, CMake, SQL / Databases, Algorithms & Data Structures, OOP, Computer Networking.
- **Spoken Languages:** Vietnamese (native), English (professional proficiency).